

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
from sklearn.linear_model import LinearRegression
warnings.simplefilter(action='ignore', category=FutureWarning)

#DATASET LOAD
df = pd.read_csv('GDP.csv')
```

1.Data Cleaning

```
In [2]: df.head(10)
```

```
Out[2]:
```

	Country	2020	2021	2022	2023	2024	2025
0	Afghanistan	20136	14278.0	14501.0	17248.0	NaN	NaN
1	Albania	15271	18086.0	19185.0	23388.0	27259.0	28372.0
2	Algeria	164774	185850.0	225709.0	247789.0	264913.0	268885.0
3	Andorra	2885	3325.0	3376.0	3786.0	4038.0	4035.0
4	Angola	66521	84375.0	142442.0	109764.0	115946.0	113343.0
5	Antigua and Barbuda	1412	1602.0	1867.0	2006.0	2225.0	2373.0
6	Argentina	385218	486040.0	632318.0	645511.0	632145.0	683533.0
7	Armenia	12642	13879.0	19514.0	24086.0	25533.0	26258.0
8	Aruba	2486	2929.0	3279.0	3649.0	3952.0	4100.0
9	Australia	1362613	1655843.0	1725461.0	1742461.0	1796805.0	1771681.0

```
In [3]: df.describe()
```

Out[3]:

	2020	2021	2022	2023	2024	2025
count	1.960000e+02	1.940000e+02	1.940000e+02	1.940000e+02	1.920000e+02	1.890000e+02
mean	4.378886e+05	5.043504e+05	5.255064e+05	5.486173e+05	5.756871e+05	5.991414e+05
std	1.942936e+06	2.220864e+06	2.347886e+06	2.453040e+06	2.573189e+06	2.693421e+06
min	5.200000e+01	6.200000e+01	6.100000e+01	6.300000e+01	6.500000e+01	6.500000e+01
25%	9.588000e+03	1.114175e+04	1.265000e+04	1.360450e+04	1.350050e+04	1.421400e+04
50%	3.533450e+04	3.771900e+04	4.156800e+04	4.363100e+04	4.713550e+04	4.782900e+04
75%	2.074810e+05	2.546132e+05	2.746308e+05	2.850355e+05	2.915108e+05	3.032930e+05
max	2.135412e+07	2.368118e+07	2.600690e+07	2.772072e+07	2.918490e+07	3.050722e+07

In [4]:

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 196 entries, 0 to 195
Data columns (total 7 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Country  196 non-null     object
1   2020     196 non-null     int64
2   2021     194 non-null     float64
3   2022     194 non-null     float64
4   2023     194 non-null     float64
5   2024     192 non-null     float64
6   2025     189 non-null     float64
dtypes: float64(5), int64(1), object(1)
memory usage: 10.8+ KB
```

In [5]:

df.isnull().sum()

```
Out[5]: Country    0
        2020      0
        2021      2
        2022      2
        2023      2
        2024      4
        2025      7
        dtype: int64
```

In [6]:

df.fillna(0 , inplace=True)

In [7]:

df.isnull().sum()

```
Out[7]: Country    0
        2020      0
        2021      0
        2022      0
        2023      0
        2024      0
        2025      0
        dtype: int64
```

```
In [8]: year_columns = ['2020', '2021', '2022', '2023', '2024', '2025']
        df[year_columns]=df[year_columns].astype(int)
```

```
In [9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 196 entries, 0 to 195
```

```
Data columns (total 7 columns):
```

```
#   Column  Non-Null Count  Dtype
---  -----  -
0   Country 196 non-null    object
1   2020     196 non-null    int64
2   2021     196 non-null    int64
3   2022     196 non-null    int64
4   2023     196 non-null    int64
5   2024     196 non-null    int64
6   2025     196 non-null    int64
```

```
dtypes: int64(6), object(1)
```

```
memory usage: 10.8+ KB
```

```
In [10]: df.head(5)
```

```
Out[10]:
```

	Country	2020	2021	2022	2023	2024	2025
0	Afghanistan	20136	14278	14501	17248	0	0
1	Albania	15271	18086	19185	23388	27259	28372
2	Algeria	164774	185850	225709	247789	264913	268885
3	Andorra	2885	3325	3376	3786	4038	4035
4	Angola	66521	84375	142442	109764	115946	113343

```
In [11]: df.duplicated().sum()
```

```
Out[11]: np.int64(0)
```

NULL Values Replace with 0 , Float Datatype - Change with Int-DataType

```
In [12]: df.head()
```

Out[12]:

	Country	2020	2021	2022	2023	2024	2025
0	Afghanistan	20136	14278	14501	17248	0	0
1	Albania	15271	18086	19185	23388	27259	28372
2	Algeria	164774	185850	225709	247789	264913	268885
3	Andorra	2885	3325	3376	3786	4038	4035
4	Angola	66521	84375	142442	109764	115946	113343

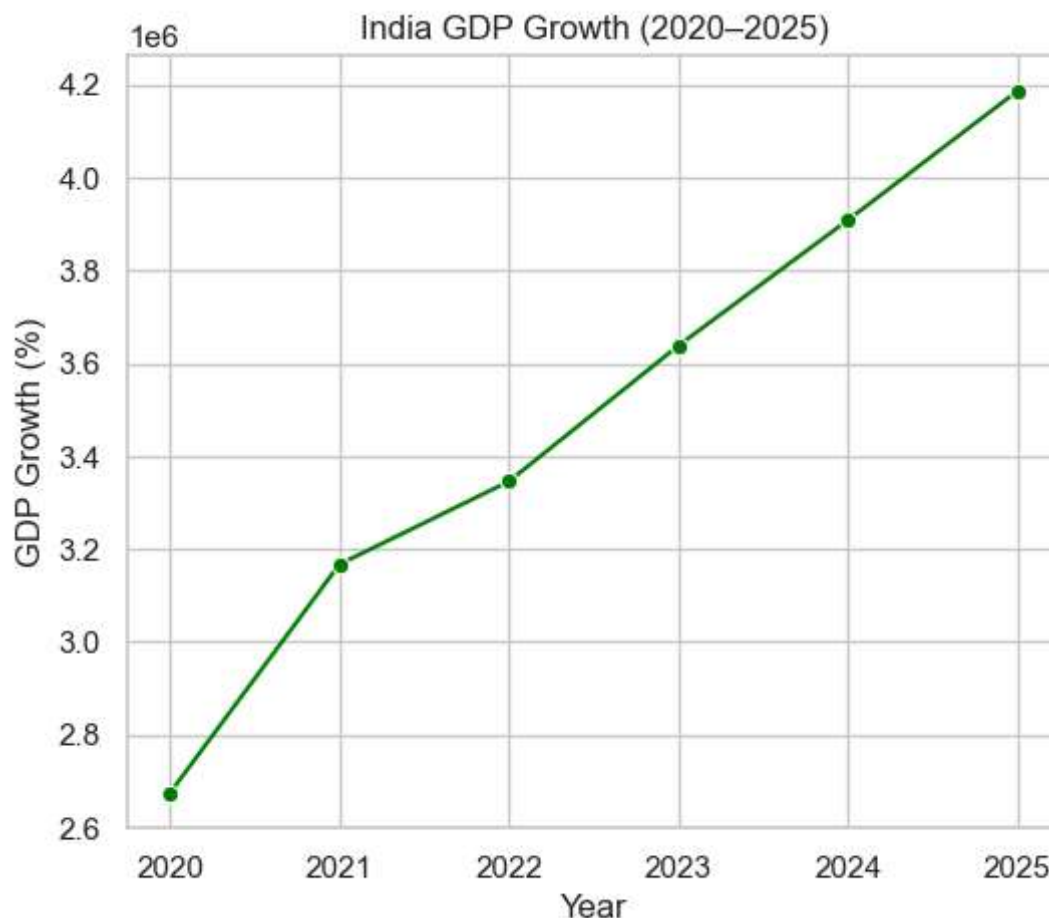
2. Trend Analysis

Plot GDP growth for a specific country from 2020–2025

```
In [13]: Country = input("Entry Specific Country Name")
Country_df = df[df['Country']== Country]

sns.set(style="whitegrid")
df_Melted = pd.melt(Country_df , id_vars=['Country'] , var_name='Year' , value_name='GDP_Growth')

plt.figure(figsize=(6,5))
sns.lineplot(x='Year', y='GDP_Growth', data=df_Melted, marker='o', color='green')
plt.title(f"{Country} GDP Growth (2020–2025)")
plt.xlabel("Year")
plt.ylabel("GDP Growth (%)")
plt.show()
```



Find the year with maximum GDP growth for each country.

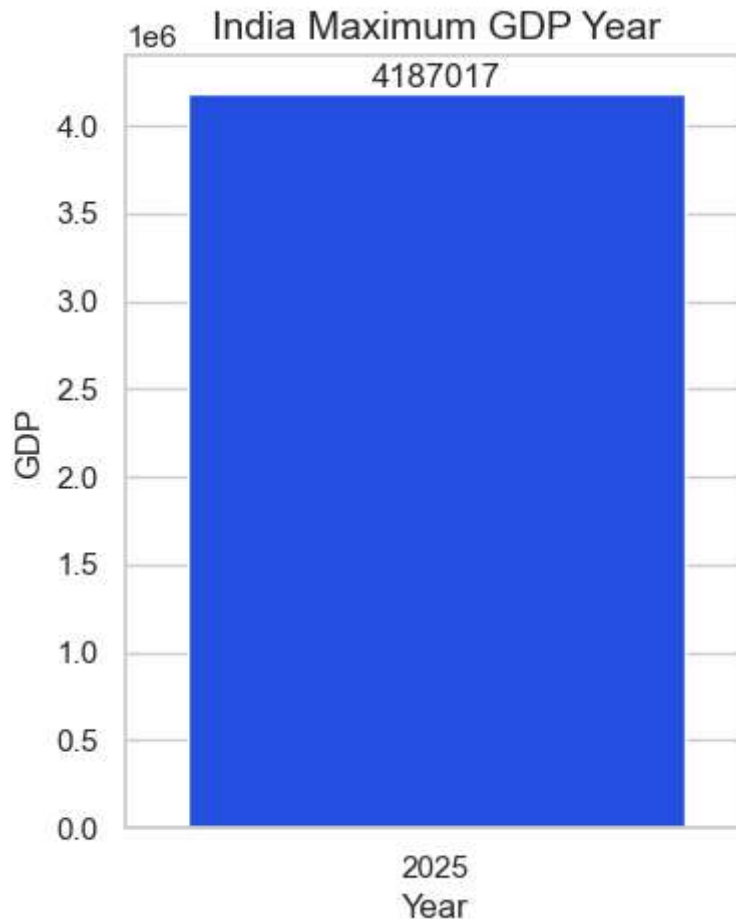
```
In [20]: Country = input("Enter the country name: ")
Country_df = df[df['Country'] == Country]
df_melted = pd.melt(Country_df, id_vars=['Country'], var_name='Year', value_name='GDP')
max_row = df_melted.loc[df_melted['GDP'].idxmax()]
max_df = pd.DataFrame([max_row])

plt.figure(figsize=(4,5))
bar = sns.barplot(
    data=max_df,
    x="Year",
    y="GDP",
    palette="bright"
)

for p in bar.patches:
    bar.annotate(
        format(p.get_height(), '.0f'),
        (p.get_x() + p.get_width()/2., p.get_height()),
        ha='center', va='bottom', fontsize=12
    )

plt.title(f"{Country} Maximum GDP Year", fontsize=14)
plt.xlabel("Year")
```

```
plt.ylabel("GDP")
plt.show()
```



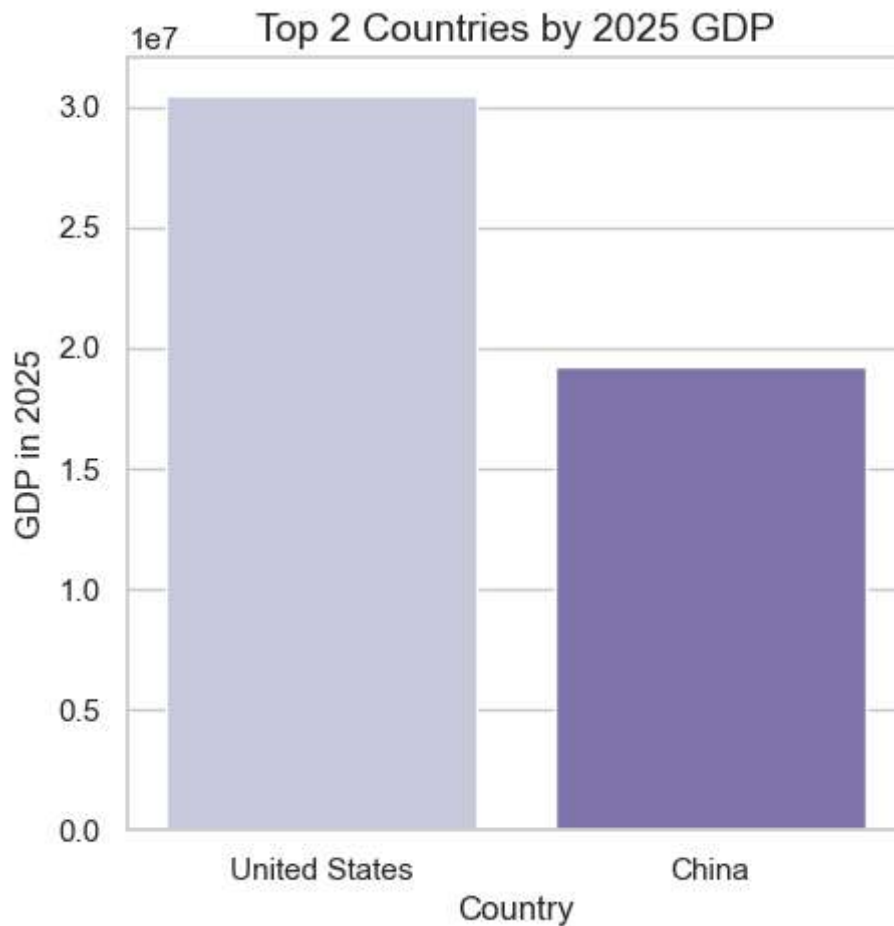
3. Comparative Analysis

Find the top 3 countries with the highest GDP in 2025.

```
In [15]: Country = int(input("Entry How many you want to see"))
Top_3 = df.sort_values(by='2025', ascending=False).head(Country)
plt.figure(figsize=(5,5))

sns.barplot(x="Country", y="2025", data=Top_3, palette="Purples")

plt.title(f"Top {Country} Countries by 2025 GDP", fontsize=14)
plt.xlabel("Country")
plt.ylabel("GDP in 2025")
plt.show()
```



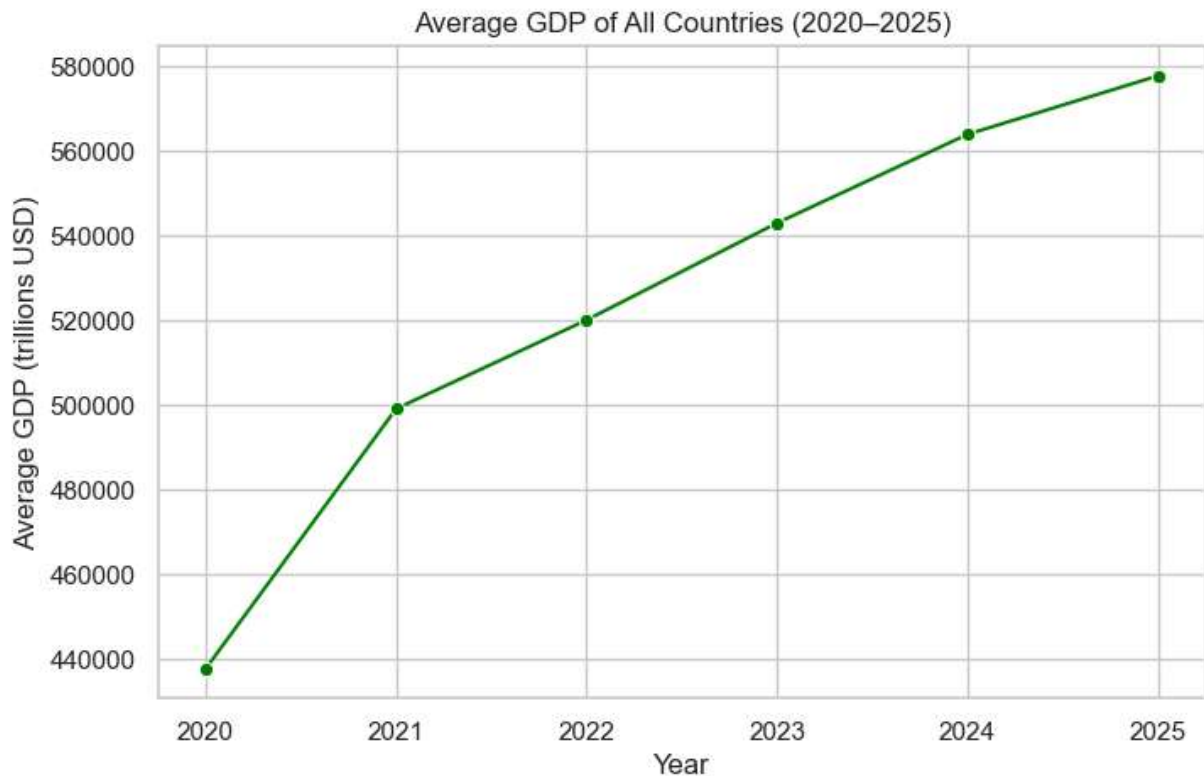
Calculate the average GDP of all countries for each year and plot it.

```
In [16]: avg_gdp = df.drop('Country', axis=1).mean()
avg_gdp = avg_gdp.reset_index()
avg_gdp.columns = ['Year', 'Average_GDP']

print(avg_gdp)

sns.set(style="whitegrid")
plt.figure(figsize=(8,5))
sns.lineplot(data=avg_gdp, x='Year', y='Average_GDP', marker='o', color='green')
plt.title("Average GDP of All Countries (2020-2025)")
plt.xlabel("Year")
plt.ylabel("Average GDP (trillions USD)")
plt.show()
```

	Year	Average_GDP
0	2020	437888.637755
1	2021	499204.005102
2	2022	520144.071429
3	2023	543019.147959
4	2024	563938.357143
5	2025	577743.540816



4. Growth Rate Analysis

Calculate the Compound Annual Growth Rate (CAGR) for each country from 2020 to 2025.

```
In [17]: Num_years = len(df.columns)-1
df['CAGR'] = ((df['2025'] / df['2020']) ** (1/Num_years)) - 1
df['CAGR'] = df['CAGR'] * 100
df_sort_Country = df.sort_values(by='CAGR', ascending=False).head(10)
print(df_sort_Country[['Country', 'CAGR']])
```

	Country	CAGR
71	Guyana	29.515136
190	Venezuela	16.754626
93	Kyrgyzstan	15.680057
72	Haiti	14.994353
63	Georgia	14.110496
28	Burundi	13.900949
69	Guinea	13.483649
102	Macau	13.209265
7	Armenia	12.955568
106	Maldives	12.417183

Predict GDP for 2026 using linear regression based on past 6 years.

```
In [18]: Top_Country = int(input("Entry Number between 1 - 196 , Showing Top Country"))

years = ['2020', '2021', '2022', '2023', '2024', '2025']
```

```

df_temp = df.assign(Total_GDP = df[years].sum(axis=1))
top5_countries = df_temp.sort_values(by='Total_GDP', ascending=False).head(Top_Coun

X = np.array([int(year) for year in years]).reshape(-1,1)

predictions = {}

sns.set(style="whitegrid")

for idx, row in top5_countries.iterrows():
    y = row[years].values
    model = LinearRegression()
    model.fit(X, y)

    gdp_2026 = model.predict(np.array([[2026]]))[0]
    predictions[row['Country']] = gdp_2026

plt.figure(figsize=(6,4))

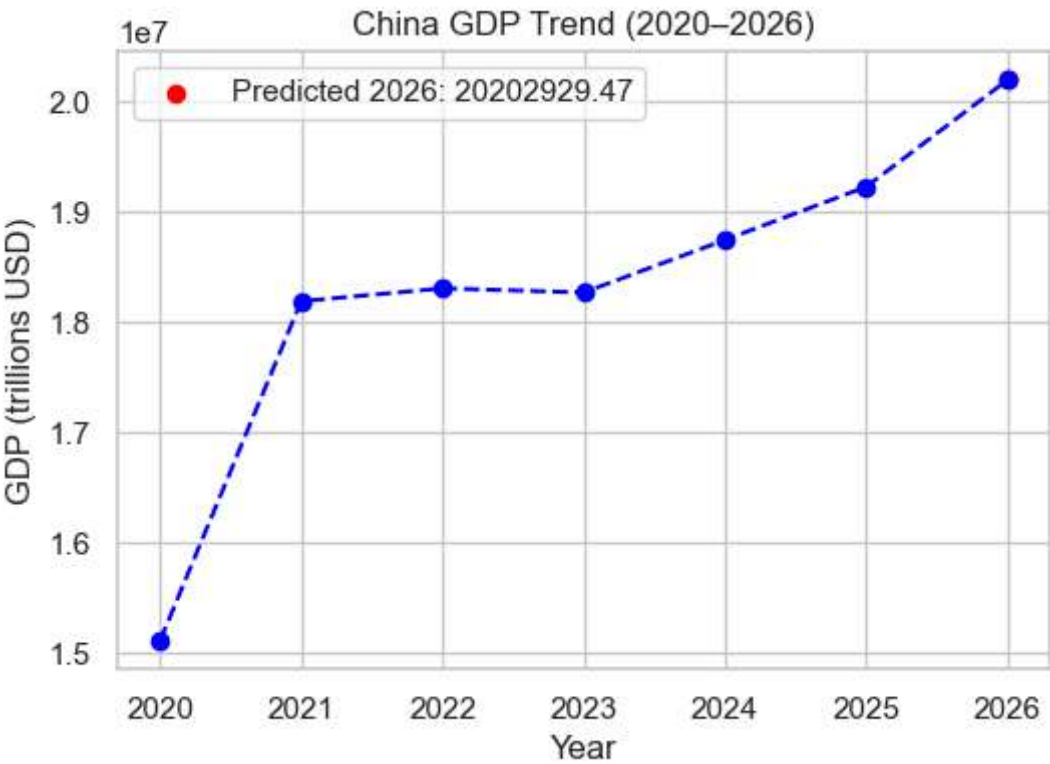
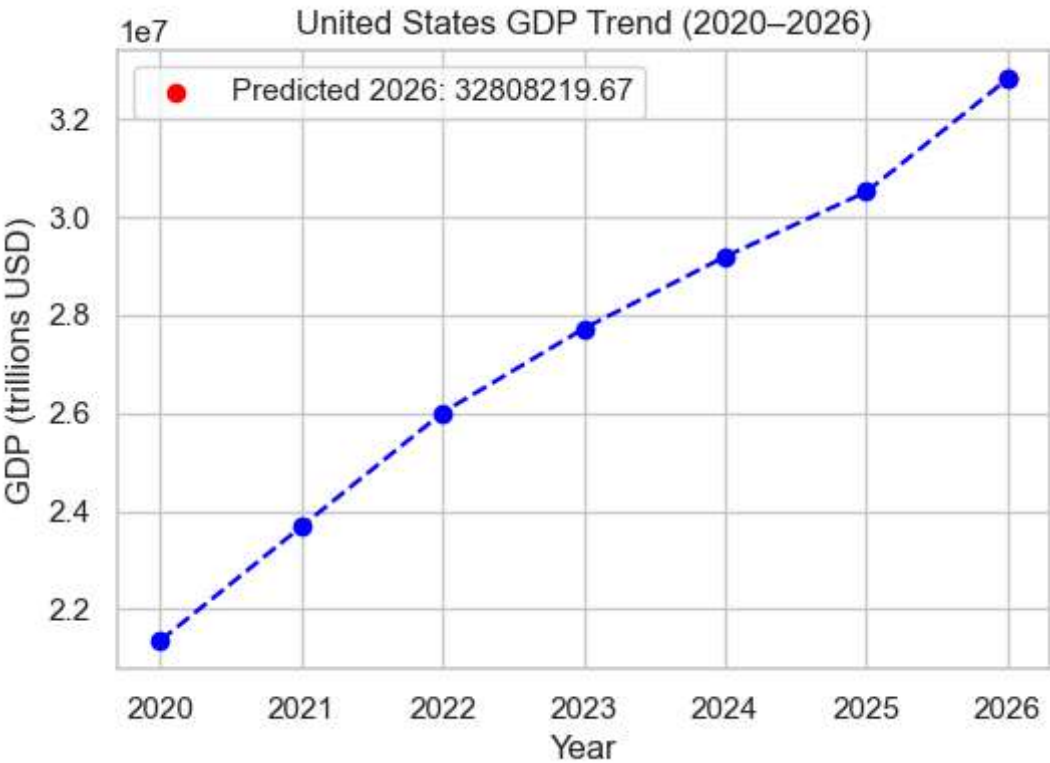
all_years = [int(y) for y in years] + [2026]
gdp_values = list(y) + [gdp_2026]

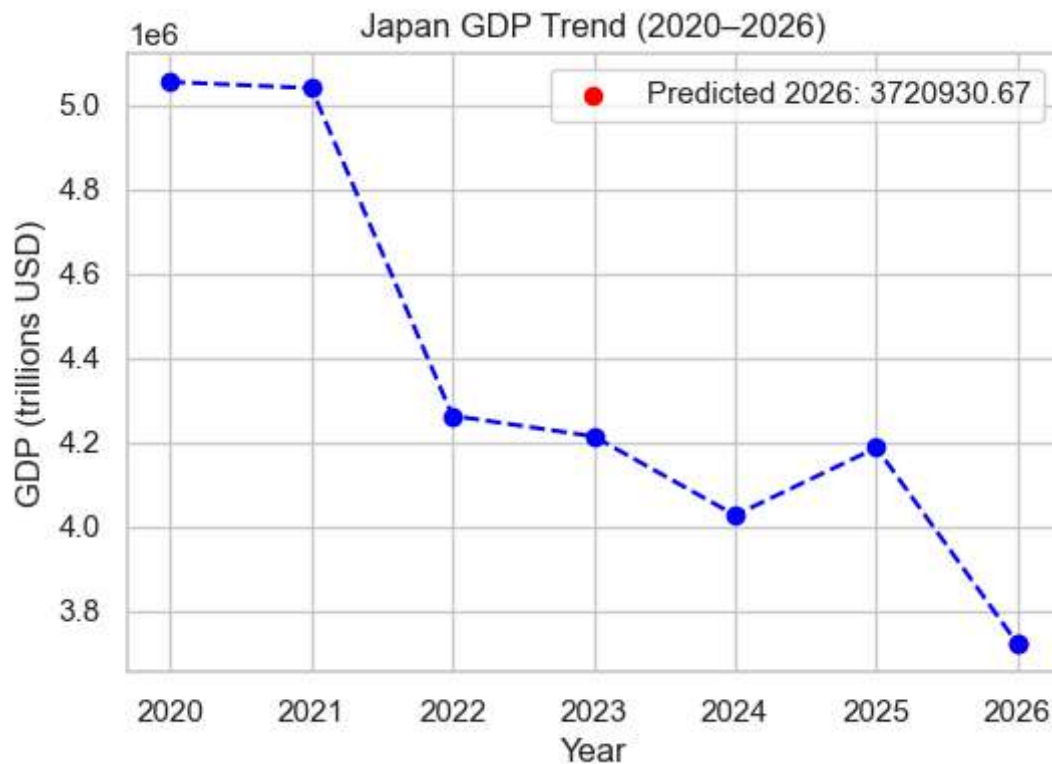
plt.plot(all_years, gdp_values, marker='o', linestyle='--', color='blue')

plt.scatter(2026, gdp_2026, color='red', label=f"Predicted 2026: {gdp_2026:.2f}")

plt.title(f"{row['Country']} GDP Trend (2020-2026)")
plt.xlabel("Year")
plt.ylabel("GDP (trillions USD)")
plt.xticks(all_years)
plt.legend()
plt.show()

```





Identify countries whose GDP is growing faster than the global average.

```
In [19]: years = ['2020', '2021', '2022', '2023', '2024', '2025']
global_avg = df[years].mean()
global_growth = global_avg['2025'] - global_avg['2020']

print("Global GDP Growth (2020–2025):", round(global_growth, 2))

df['GDP_Growth'] = df['2025'] - df['2020']

faster_countries = df[df['GDP_Growth'] > global_growth]
print("\n Countries whose GDP grew faster than global average:\n")
print(faster_countries[['Country', 'GDP_Growth']])
```

Global GDP Growth (2020-2025): 139854.9

Countries whose GDP grew faster than global average:

	Country	GDP_Growth
6	Argentina	298315
9	Australia	409068
17	Belgium	155595
24	Brazil	649866
32	Canada	569656
36	China	4128348
37	Colombia	157418
60	France	565486
64	Germany	807815
77	India	1512165
78	Indonesia	370688
79	Iran	145485
81	Ireland	162635
82	Israel	172590
83	Italy	516899
112	Mexico	571575
123	Netherlands	340197
138	Poland	374030
142	Romania	151023
143	Russia	588278
148	Saudi Arabia	349478
153	Singapore	215608
160	Spain	510760
168	Switzerland	205837
179	Turkey	720292
184	United Arab Emirates	199125
185	United Kingdom	1140475
186	United States	9153092
191	Vietnam	144660

In []: